ENCE 3501 VLSI

Lab 3: NMOS with Basic ESD Pad Protection

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1. Introduction

The aim of this lab is to design a die that allows an NMOS transistor to be connected to the outside world, where every pad of the padframe has a basic ESD protection. The work was done in Electric VLSI with the MOSIS CMOS technology and the C5 models. The NMOS cell from Tutorial 3 and the pad cell from the previous lab were taken as the starting point. Firstly, the two protection diodes were designed, then these diodes were used to make the pad cell with ESD protection. After that, the pad cells were placed in a square padframe with the VDD and GND bus, and at the end the NMOS was placed inside the frame and wired to the pads. The schematic and the layout of each step are explained below, and the design rule check results are given at the end of the report.

2. Schematic

2.1 pWell-nActive and pActive-nWell diodes

Two diodes are needed for the protection. The first one is the pActive-nWell diode, where the p+ active region is the anode and the n-well is the cathode, so the ports of this cell are named pActive and nWell. The second one is the pWell-nActive diode, where the p-well is the anode and the n+ active region is the cathode, so its ports are named pWell and nActive. Both cells are drawn with the diode symbol of the schematic library, and the arrow points from the p side to the n side. Off-page connectors are used at both ends and these become the ports of the cell. I kept the port names same as the layout ports, so that the schematic and the layout match each other. An icon view was also made for each diode, so that they can be placed as blocks inside the pad cell.

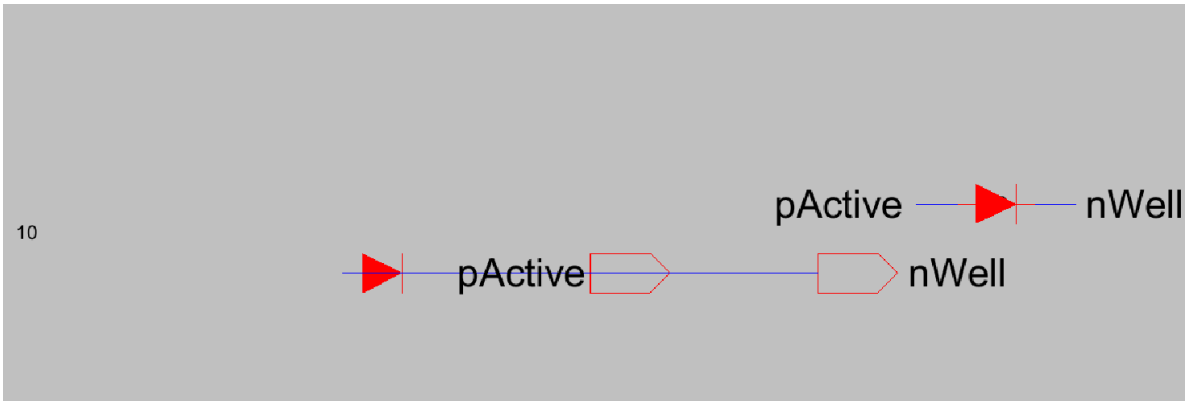


Figure 1: Schematic and icon of the pActive-nWell diode

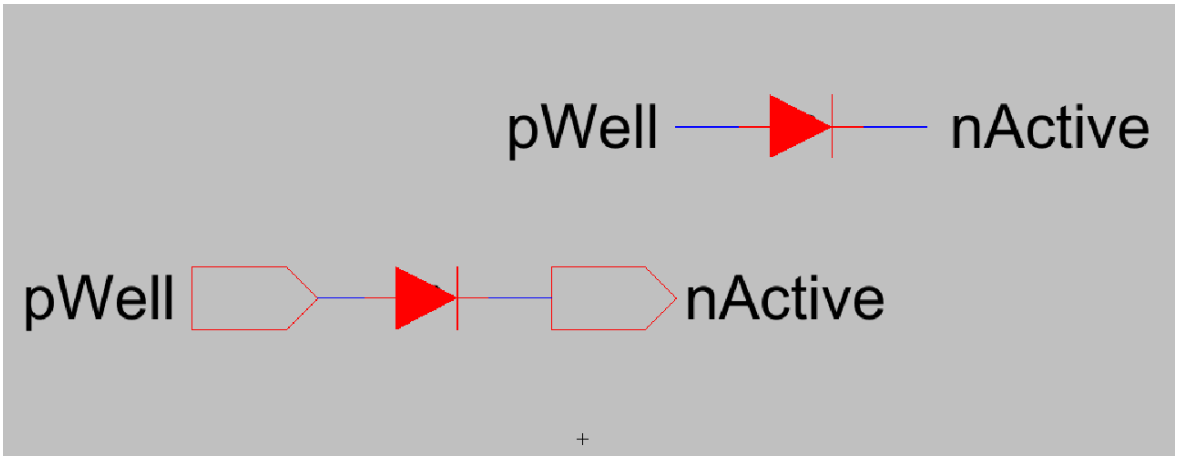


Figure 2: Schematic and icon of the pWell-nActive diode

2.2 Pad cell with ESD protection

In the pad_esd schematic the inout node is connected to the pActive port of the pActive-nWell diode and to the nActive port of the pWell-nActive diode. The nWell port of the first diode goes to vdd and the pWell port of the second diode goes to gnd. During normal operation the pad voltage stays between gnd and vdd, so both diodes are reverse biased and they take no current. For this reason the signal goes to the NMOS without any change. However, if a static discharge pushes the pad about 0.7 V above vdd, the upper diode turns on and sends the charge to the vdd rail. In the same way, if the pad is pulled about 0.7 V below gnd, the lower diode conducts and the current comes from the gnd rail. In this way the pad voltage is clamped inside a safe range and the thin gate oxide of the transistor is not damaged. The cell has three ports, which are inout, vdd and gnd, and an icon was made for it so it can be reused in the padframe.

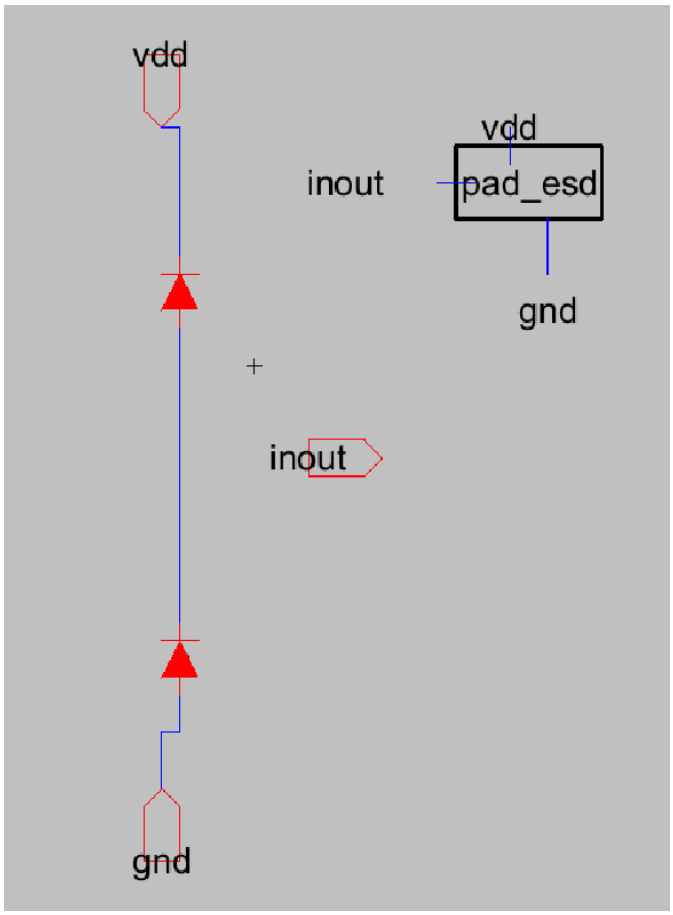


Figure 3: Schematic and icon of the pad cell with ESD protection (pad_esd)

2.3 Padframe with the VDD and GND bus

The padframe schematic uses the pad_esd cell with its inout port connected to the bus pin[1:8], which stands for the eight pads of the frame. The vdd and gnd ports of the pad cell go directly to the vdd and gnd ports of the padframe, so all the pads share one VDD net and one GND net. I connected them together because the diodes of every pad must clamp to the same supply levels. If each pad had its own supply, one pad could clamp at a different voltage than the others, and the protection would not be reliable. In the layout this schematic is realised with eight copies of the pad cell.

About the number of pads, the padframe has to be a square. In the general idea given with the lab, the frame has two pads on each side, so I used eight pads. From these, only four are connected to the NMOS, which are pin[1], pin[4], pin[7] and pin[8], because the transistor has four terminals. The other four pads (pin[2], pin[3], pin[5] and pin[6]) are not used, but they keep the frame square and symmetric.

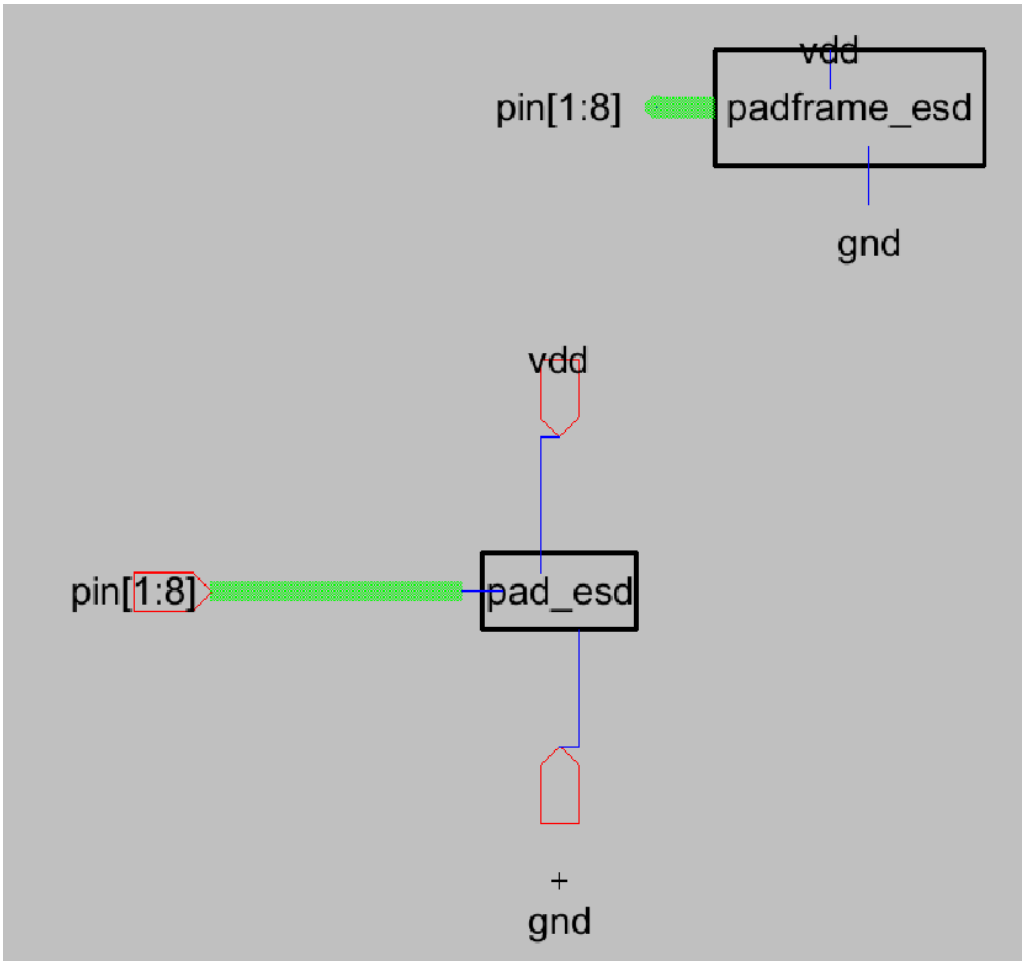


Figure 4: Schematic and icon of the padframe with the VDD and GND bus (padframe_esd)

2.4 Padframe with the NMOS

In the final_ic_esd schematic the padframe_esd icon is placed with its pin[1:8] bus, and the NMOS icon is wired to the single pins of this bus. Pin[1] is connected to the drain, pin[4] to the gate, pin[8] to the source and pin[7] to the bulk. The pins were chosen to follow the position of the pads in the layout, where pin[1] and pin[8] are on the right side of the frame, pin[4] is on the left side and pin[7] is at the bottom. The vdd and gnd ports of the padframe stay available for the supply connection, and the four pads which are not needed are left open.

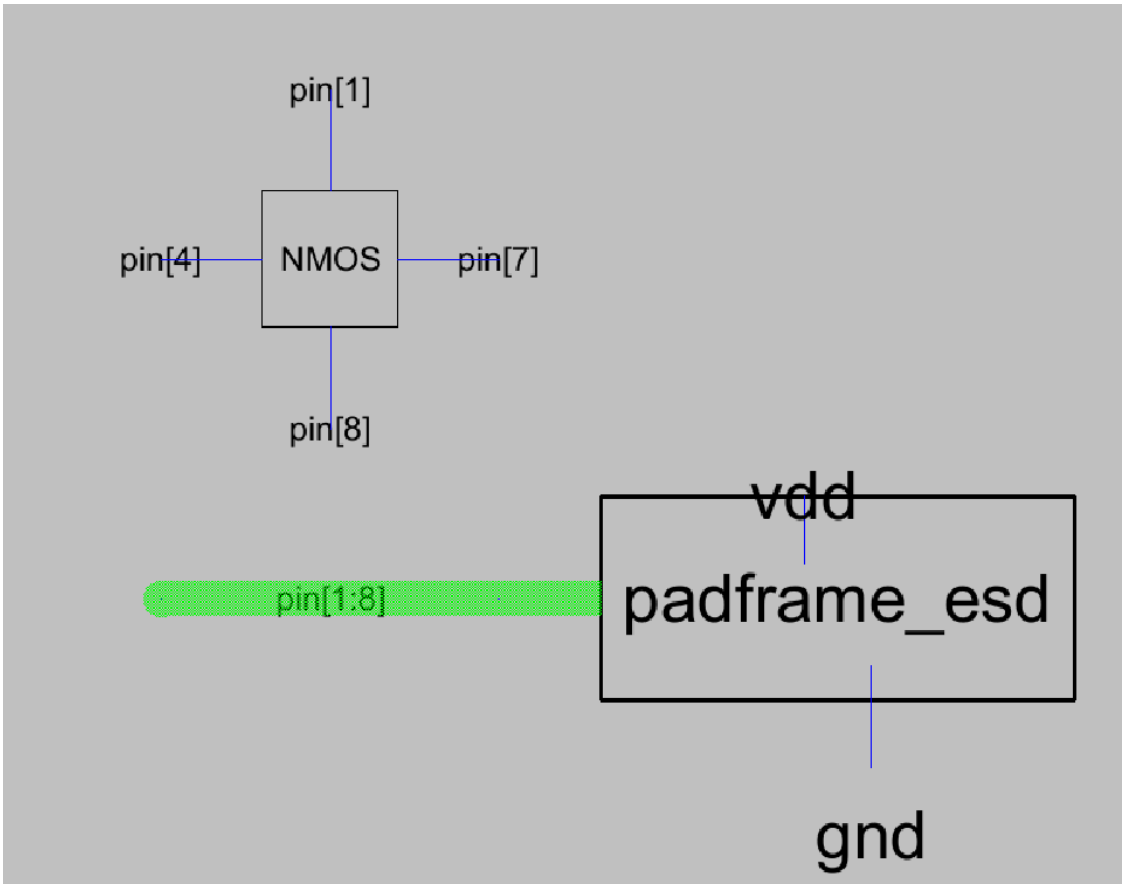


Figure 5: Schematic of the padframe with the NMOS (final_ic_esd)

3. Layout

3.1 pWell-nActive and pActive-nWell diodes

For the pActive-nWell diode a Metal-1 to P-Active contact is placed and it is surrounded by an n-well, which makes the p+ region of the diode. A Metal-1 to N-Well contact is placed next to it to hold the n-well, and this is the cathode. For the pWell-nActive diode a Metal-1 to N-Active contact is placed in the p-well and a Metal-1 to P-Well contact is placed next to it. Large contact arrays were used in both diodes so that they can carry a lot of current during an ESD event. The two Metal-1 ends of each cell are exported with the same names as in the schematic.

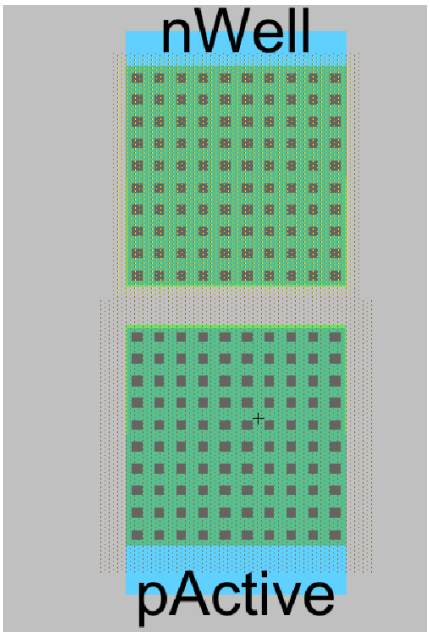


Figure 6: Layout of the pActive-nWell diode

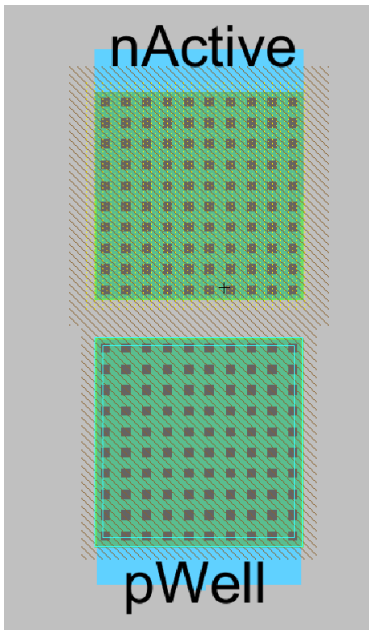


Figure 7: Layout of the pWell-nActive diode

3.2 Pad cell with ESD protection

The pad cell from the previous lab gives the pad opening, and its inout export is on Metal-2. Both diodes are placed below the pad. The pWell-nActive diode is on the left, with its nActive end connected to the pad through Metal-1 and a Metal1-Metal2 contact. Its pWell end goes to the gnd rail by a Metal-2 strap, which crosses over the vdd rail without any contact, so the two rails are not shorted. The pActive-nWell diode is on the right and is rotated by 180 degrees, so that its pActive end goes up to the pad and its nWell end goes down to the vdd rail. The vdd and gnd rails are two parallel Metal-1 bars under the diodes, and they are exported as vdd and gnd. The spacing between the rails and between the contacts was kept above the Metal-1 spacing rule.

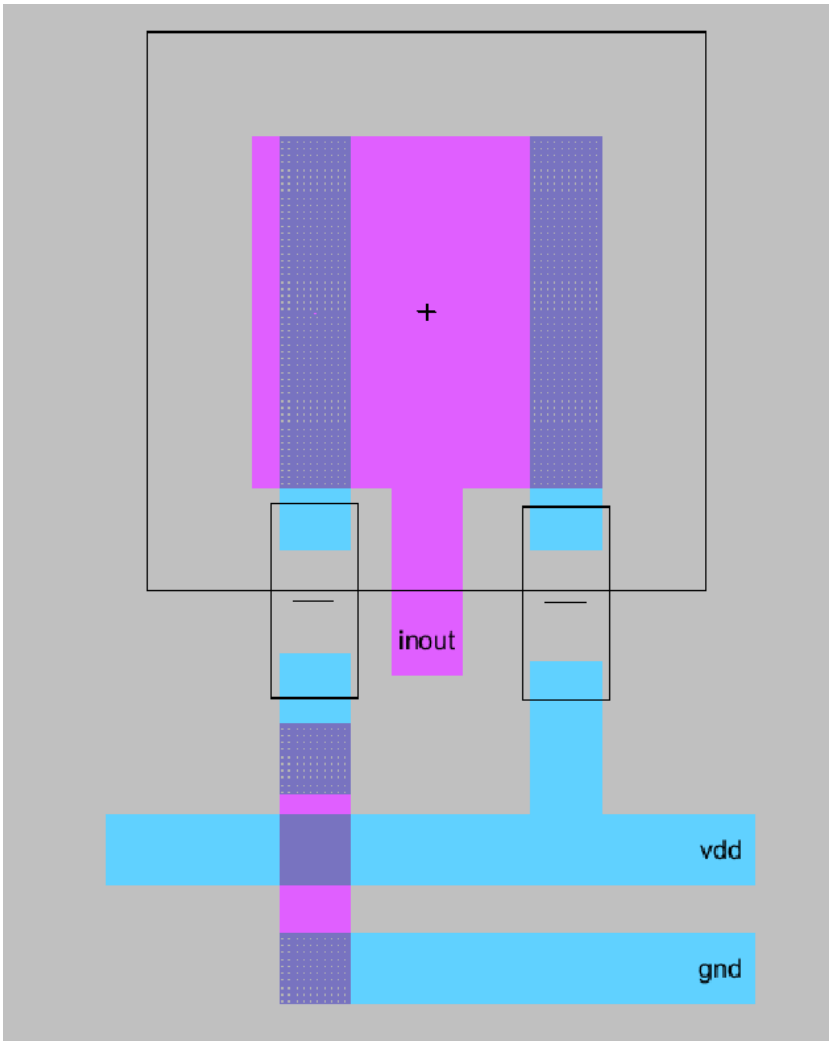


Figure 8: Layout of the pad cell with ESD protection (pad_esd)

3.3 Padframe

The padframe layout is made with eight pad_esd cells in a square ring, with two pads on each side, and the pads are rotated so that their rails face the inside of the frame. The vdd rails make the outer ring and the gnd rails make the inner ring, and both are Metal-1 wires joined at the corners and between the neighbouring pads. The inout port of each pad is exported as pin[1] to pin[8], and the supply rails are exported as vdd and gnd. Two things were corrected during the design. A left over Metal-1 stub at the end of the gnd rail inside the pad cell was removed because it was overlapping the bus wires of the neighbouring pad. Further, some bus wires were first running too close to the pWell contact of the next pad, so they were redrawn along the centre of the rails.

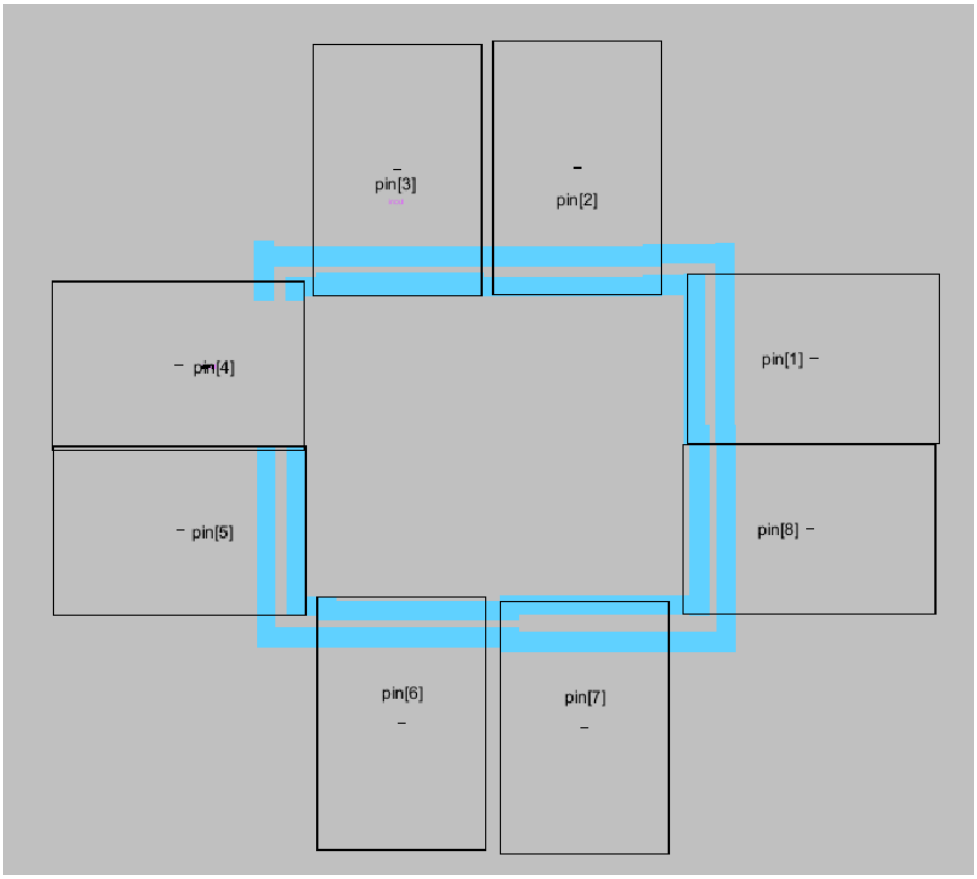


Figure 9: Layout of the padframe with the VDD and GND bus (padframe_esd)

3.4 Padframe with the NMOS

The NMOS layout has an n-transistor with an N-Active contact on each side for the source and the drain, and both of them are joined to the diffusion with N-Active arcs. The poly gate goes up to a Metal-1 to Poly contact, and a P-Well contact is placed below the transistor for the bulk. This contact sits in the same p-well as the transistor, so no extra wire is needed inside the cell. The four Metal-1 ends are exported as g, s, d and b.

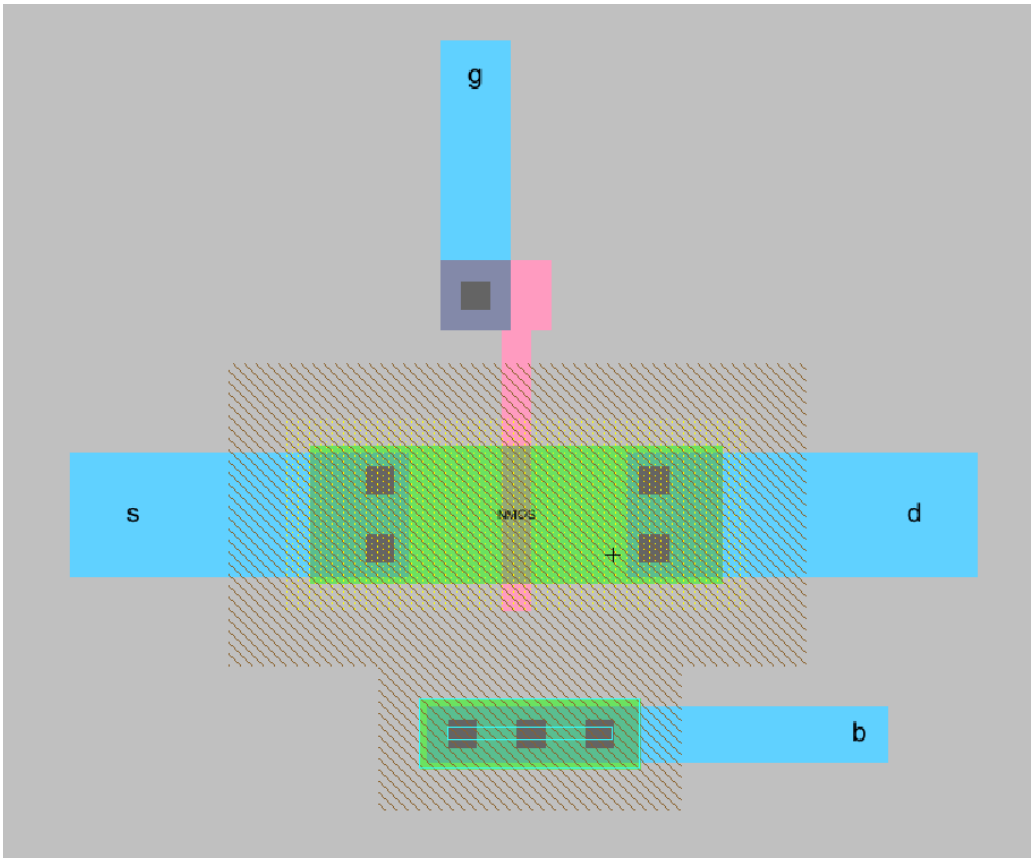


Figure 10: Layout of the NMOS with exports g, s, d and b

In the final layout the NMOS is placed in the middle of the padframe. Each terminal leaves its port with a short Metal-1 stub of 4 lambda width, then a Metal1-Metal2 contact is used and the long run to the pad is done on Metal-2. Pin[4] goes to the gate, pin[1] to the drain and pin[7] to the bulk. Pin[8] has to reach the source on the left side, so its Metal-2 wire goes around below the transistor. It crosses over the Metal-1 stub of the bulk, which is allowed because they are on different layers. The Metal-2 wires never cross each other, so no short is made between the pads. All the contacts were placed away from the rails of the padframe to avoid spacing errors.

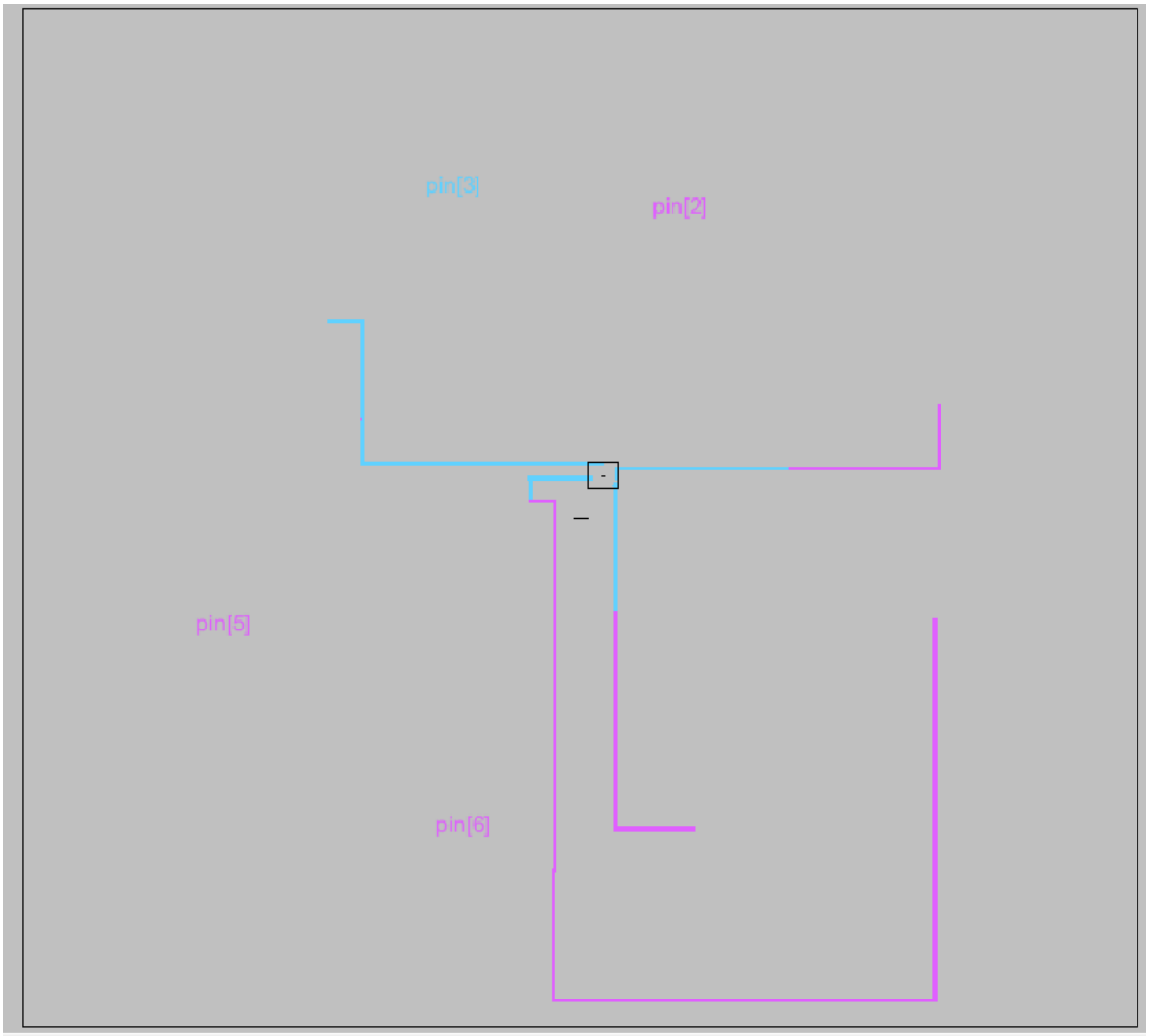


Figure 11: Layout of the padframe with the NMOS inside (final_ic_esd)

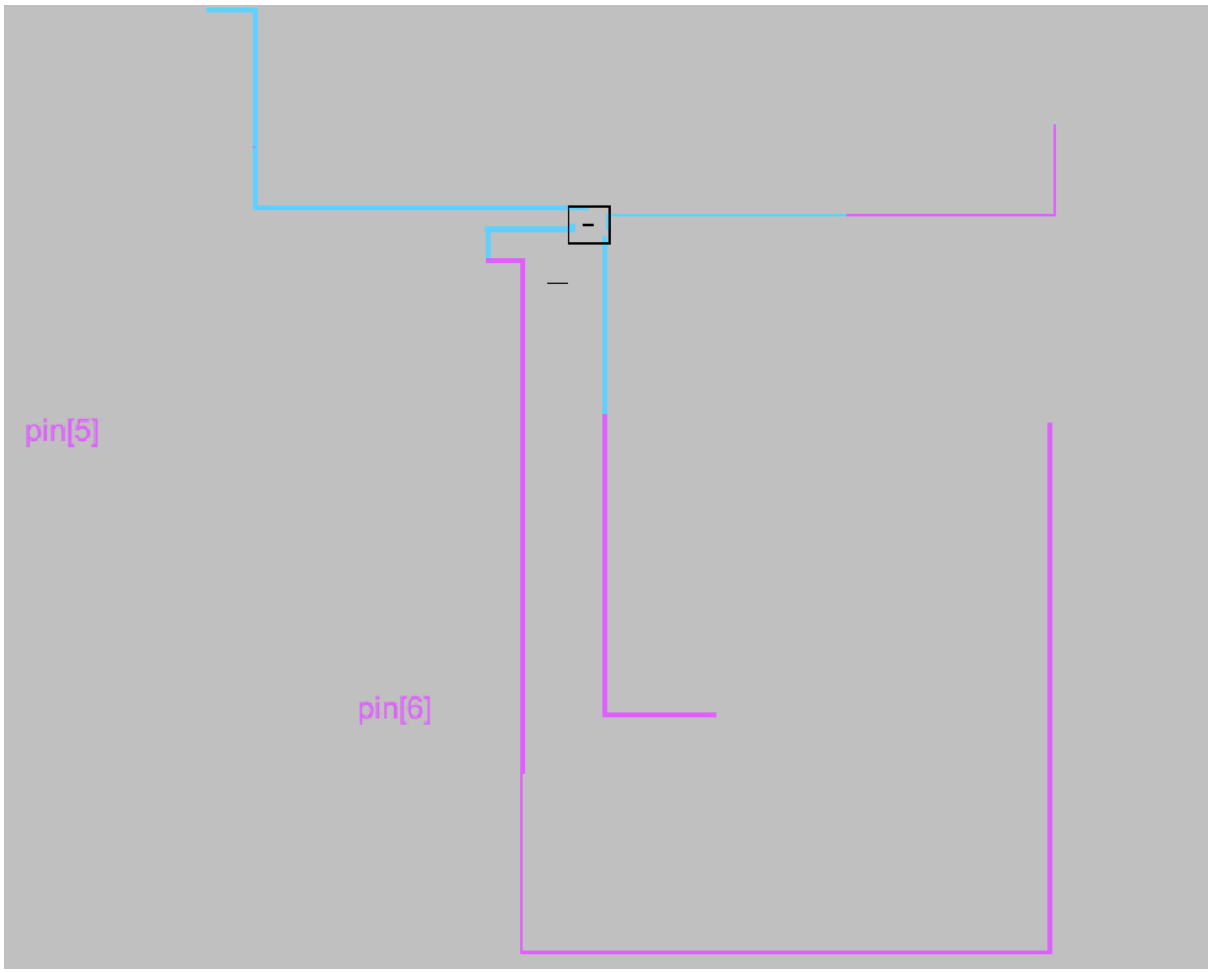


Figure 12: Connections of the four pads to the NMOS terminals

4. Design Rule Check

The design rule check of Electric was run with the Mosis rules on the main layout cells. The diodes are checked as part of these cells, because they are placed inside the pad cell. For pad_esd, padframe_esd and final_ic_esd the result was 0 errors and 0 warnings, and the same result was found for the NMOS layout. The messages are shown below.

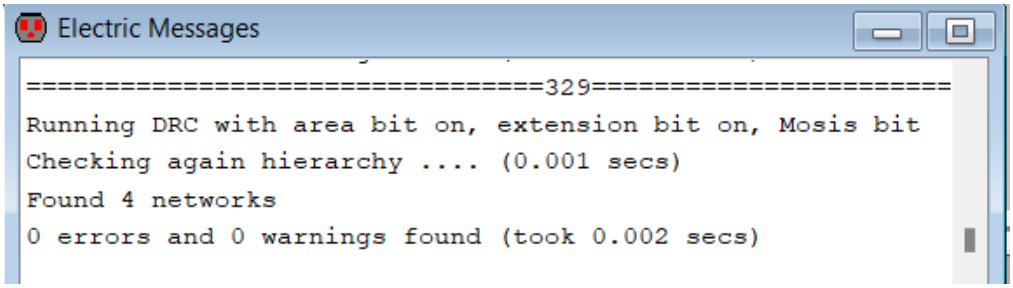


Figure 13: DRC result for pad_esd

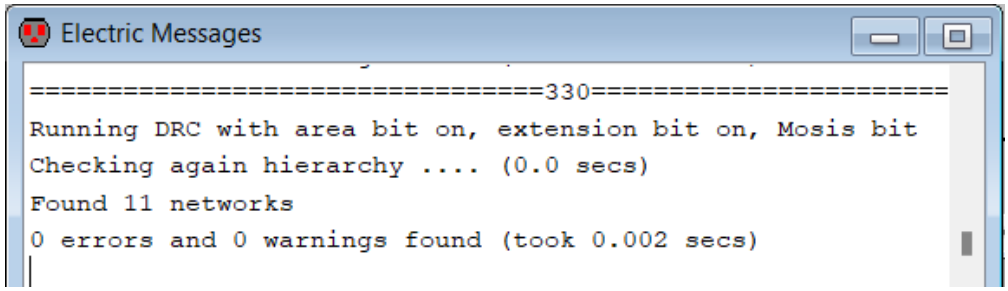


Figure 14: DRC result for padframe_esd

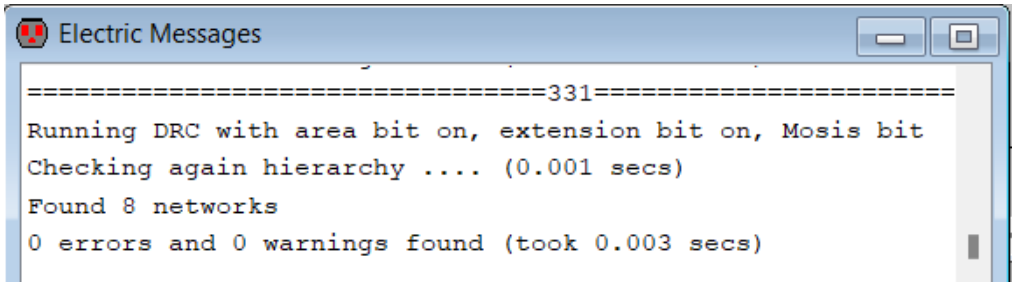


Figure 15: DRC result for final_ic_esd

5. Conclusion

To conclude, the lab gave a complete flow from two diodes to a padframe with an NMOS inside it. The two diodes in each pad protect the transistor in both directions, because one clamps the pad to vdd and the other to gnd. Joining the vdd and gnd of all pads in one bus keeps the clamping level same for every pad. In my view, the most important part of the layout was the routing, since the wires of different pads have to stay on the correct layer and never touch the rails or the contacts of the other pads. In nutshell, the final design has eight protected pads, four of them wired to the drain, gate, source and bulk of the NMOS, and it passes the design rule check without any error.